

ASSIGNMENT-I

Name: C. Vaishnavi

ID No: B210350

Roll no: 15

Class: CSE C4

- 1) A company surveyed 100 customers and found that the average satisfaction rating (on a scale of 1 to 5) was 4.2, with a standard deviation of 0.6 construct a 95% confidence interval for the true mean satisfaction score of all customers.

Given that $\bar{X} = 4.2$ $\sigma = 0.6$ $z_{\alpha/2} = 95\% = 1.96, n = 100$ confidence Interval = $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$CI = 4.2 \pm 1.96 \times \frac{0.6}{\sqrt{100}}$$

$$= 4.2 \pm 0.096$$

$$CI = (4.104, 4.296)$$

∴ So, we are 95% confident that the true mean satisfaction score of all customers lies between 4.104 and 4.296.

- 2) A hospital collects data from a random sample of 40 patients. The average length of stay is 5.6 days, with a standard deviation of 1.8 days. Estimate 99.9% Confidence interval for the average hospital stay for all patients.

Given that $\bar{x} = 5.6$

$$\sigma = 1.8$$

$$n = 40$$

$$Z_{\alpha/2} = 99\% = 2.516$$

$$CI = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$= 5.6 \pm 2.516 \times \frac{1.8}{\sqrt{40}}$$

$$= 5.6 \pm 2.516 \times 0.2844$$

$$= 5.6 \pm 0.7326$$

$$CI = (4.867, 6.333)$$

So, we are 99% confident that the average hospital stay for all Patients lies between 4.867 and 6.333 days.

3)

Student scores Analysis

Scores of 10 students in a class on a test:

[45, 56, 67, 89, 90, 66, 75, 88, 91, 60]

- Compute mean, median, and standard deviation
- Create a frequency distribution with class intervals (e.g.: 40-50, 50-60 etc)
- Plot a histogram and comment on the skewness of the distribution.

Given data in ascending order

45, 56, 60, 66, 67, 75, 88, 89, 90, 91

$$a) \text{ Mean} = \frac{45+56+60+66+67+75+88+89+90+91}{10}$$

$$= \frac{727}{10}$$

$$\text{mean} = 72.7$$

$$\text{Median} = \left(\frac{\frac{n}{2} + \frac{(n+1)}{2}}{2} \right) = \frac{67+75}{2} = 71$$

$10/2 = 5^{\text{th}}$ value

$12/2 = 6^{\text{th}}$ value

Standard deviation:

$$(45 - 72.7)^2 = 771.84$$

$$(75 - 72.7)^2 = 5.29$$

$$(56 - 72.7)^2 = 282.24$$

$$(88 - 72.7)^2 = 234.09$$

$$(60 - 72.7)^2 = 163.84$$

$$(89 - 72.7)^2 = 263.17$$

$$(66 - 72.7)^2 = 44.89$$

$$(90 - 72.7)^2 = 299.29$$

$$(67 - 72.7)^2 = 32.49$$

$$(91 - 72.7)^2 = 330.62$$

$$\begin{aligned} \sum (x - \bar{x})^2 &= 771.84 + 282.24 + 163.84 + 44.89 + 32.49 + \\ &\quad 5.29 + 234.09 + 263.17 + 299.29 + 330.62 \\ &= 2427.86 \end{aligned}$$

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

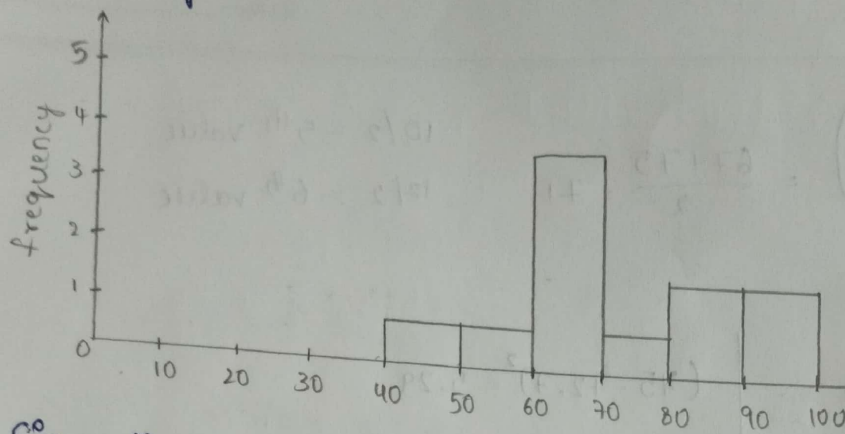
$$\sigma = \sqrt{\frac{2427.86}{10}} = 15.58$$

$$\boxed{\sigma = 15.58}$$

b) class-interval frequency.

40 - 50	1
50 - 60	1
60 - 70	3
70 - 80	1
80 - 90	2
90 - 100	2

c) Histogram



Since the data is concentrated toward the lower scores the distribution appears to be slightly negatively skewed.

- 4) A hospital collected the ages of 20 patients, visiting the outpatient department.
- [22, 34, 45, 29, 31, 39, 41, 36, 27, 40, 33, 35, 38, 37, 28, 30, 32, 60, 55, 58]
- Find the mean, median and range of patient ages.
 - Determine the interquartile range.
 - Check if the data is positively or negatively skewed.

sol: Given data

22, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 45, 55, 58, 60

$$\begin{aligned} \text{i) Mean} &= \frac{22+27+28+29+30+31+32+33+34+35+36+37+38+39+40+41+45+55+58+60}{20} \\ &= \frac{750}{20} = 37.5 \end{aligned}$$

$$\text{Median} = \frac{\left(\frac{20}{2}\right)^{\text{th}} + \left(\frac{22}{2}\right)^{\text{th}}}{2} = \frac{10^{\text{th}} + 11^{\text{th}}}{2} = \frac{35+36}{2} = 35.5$$

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$= 60 - 22$$

$$= 38$$

$$ii) Q_{25\%} = \frac{25}{100} \times (20+1)$$

$$= 0.25 \times 21$$

$$= 5.25$$

$$Q_{75\%} = \frac{75}{100} \times (20+1)$$

$$= 0.75 \times 21$$

$$= 15.75$$

$$Q = Q_{75\%} - Q_{25\%}$$

$$Q = 15.75 - 5.25$$

$$Q = 10.5$$

$$iii) \text{Mean} = 37.5$$

$$\text{Median} = 35.5$$

Since $\text{Mean} > \text{Median}$, the data is positively skewed
(right-skewed) //